Software Requirement Specifications
for
AUC Rideshare

Version 1.0

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Introduction

1.1 Purpose

AUC Rideshare is an application designed to facilitate carpooling among AUC students, faculty, and staff. The app aims to connect members of the AUC community, allowing them to request rides to AUC with fellow AUCians. Our goal is to reduce transportation costs, provide safer ride options, and foster a sense of community by linking users with similar schedules and preferences. AUC Rideshare allows users to schedule rides in advance or in real-time, share ride details, and communicate seamlessly.

1.2 Stakeholders

Primary Stakeholders:

- AUC Students: Primary users who will use the app to find rides to and from AUC.
- AUC Faculty & Staff: Users who may also use the app for commuting.
- App Developers: Responsible for building and maintaining the app.

Secondary Stakeholders:

- AUC Administration: May provide support and integration with AUC's transportation systems.
- Competitors: Other rideshare services like Uber, Careem, and inDrive.
- Local Transportation Authority: May regulate or provide support for the app.

1.3 User Narrative

Sarah is a sophomore at AUC who lives off-campus and doesn't own a car. She usually orders an Uber to campus but often struggles to find rides especially to her early morning classes and service quality is not always the best. She downloads the AUC Rideshare app and registers using her AUC ID and phone number. During registration, she selects her preferences, such as preferring female drivers, enjoying pop music, and being interested in discussing movies and TV shows. She also inputs her class schedule for the week and her adress.

Laila, a junior at AUC, also downloads the AUC Rideshare app but registers using her AUC ID and phone number as a driver. During registration, she provides details about

her car, including the model, make, color, and license plate. She also sets her preferences and inputs her class schedule for the week along with her address.

The app recommends Laila to Sara, and Sarah schedules a ride with Laila for her 8:30 AM class and Laila confirms the ride. On the day of the ride, Sarah receives a notification when Laila is nearby, and they can communicate via their phone numbers to coordinate the pickup. Sarah pays for the ride using Telda, one of the integrated payment options, and both receive a confirmation of the transaction. After a few rides, Sarah adds Laila as a friend on the app, building a new friendship and making it easier to schedule future rides.

1.4 References

TBD

Product Overview

2.1 Product Perspective

2.1.1 Interfaces

User Interfaces:

- **Registration:** Home screen where users can register and login as a user or driver.
- **Ride Ordering Screen:** For users to order rides in real time.
- **Ride Scheduling Screen:** For riders to schedule rides in advance.
- **Friends Tab:** For users to connect with others and view their schedules and preferences.
- **Payment Screen:** For riders to select payment methods (e.g., Telda, Instapay, Vodafone Cash) and confirm transactions.
- Due to time constraints, more features will not be available (e.g., chat box)

System Interfaces:

- AUC Database (For user authentication)
- Location Tracking API
- Payment Gateways

2.1.2 Site Adaptation Requirements

TBD

2.2 Product Functions

2.3 User Characteristics

- **Students:** Main users who will use the app to find rides to and from AUC.
- **Drivers:** Users who offer rides and share their vehicles with others.
- **Admins:** Responsible for managing the system, handling reports, and ensuring the app runs smoothly.

The use case diagram containing more information about what each user can do is available in the design document.

2.4 Limitations

- **Time Constraints:** We have a limited timeframe (3 months) to develop a working prototype, which may restrict the scope of features.
- **Team Expertise:** Limited experience may slow down development and require additional learning.
- **Third-Party API Reliability:** Dependence on external APIs for location tracking may introduce delays or inconsistencies in functionality.
- **Testing Limitations:** Limited resources for extensive testing may result performance issues in the initial release.

2.5 Assumptions and Dependencies

- **AUC Database:** We will be assuming that the app will have access to AUC's student database in order to authenticate each user registering for the app using their AUC ID.
- **Third-Party API:** We will be depending on a third party API for location tracking and assuming it functions reliably.
- **Payment Gateways:** We will be depending on payment gateways such as Telda, Instapay, and Vodafone Cash.

2.6 Apportioning of Requirements

TBD

Specific Requirements

3.1 Registration

Functional:

- Users should register with their AUC ID and phone numbers.
- Drivers should register their car information like its model, make, color, and license plate along with their AUC ID and phone numbers as well.
- Users should choose preferences during registration (e.g., driver gender, music taste, hobbies, topics to talk about, etc.).
- Users should be able to input their class schedules for the week during registration.

Nonfunctional:

- Authentication should be done using AUC's database.

3.2 Ride Details

Functional:

- Riders should be able to schedule rides in advance.
- Riders should be able to schedule rides in real-time.
- Riders and drivers should see each other's phone numbers when a ride is confirmed to facilitate communication.
- There will be penalties for:
    - Drivers who are late to pick up riders.
    - Riders who are late to meet drivers.
- Drivers and riders will share their location so they can find each other.

Nonfunctional:

- Location sharing shall be real-time and accurate through a third party API.
- Ride request should time out after searching for a driver for 5 minutes without a response
- Riders cannot book a ride more than 3 days in advance

3.3 Friends Feature

Functional:

- There will be a friends screen where riders and drivers can view each other's schedules and preferences.
- Users should be able to "friend" other users with similar schedules or preferences to have a go-to ride partner.

Nonfunctional:

TBD

3.4 Payment

Functional:

- Riders shall be able to pay using cash, Telda, Instapay, or Vodafone Cash.

Nonfunctional:

- Payment confirmation shall be required for each transaction.

3.5 safety and reporting

Functional:

- Users shall be able to send an alert to the AUC Security office as they monitor live location and ride details.
- Users shall be able to report unsafe behavior in the app.
    - Reports shall be sent to admins for review, and users who receive multiple reports will be flagged.
    - Admins shall be able to suspend users who violate safety rules.
    - The system shall store all ride data automatically such as location and driver's information

Nonfunctional:

- Emergency alerts shall be processed within 3 seconds.
- Reports shall be reviewed within 48 hours

3.6 Ride review and rating

Functional:

- Riders and drivers shall rate each other after each ride on a 5 star scale.
- Users shall be able to write feedback.
- Users and drivers will have a rating displayed on their profiles.
- Admins shall be able to review flagged users.

Nonfunctional:

- Ratings shall update in real time after each ride.
- Any user with a rating below 2 stars shall be flagged.

3.7 ride cancellation and refund

Functional:

- Users shall be able to cancel a ride before the driver arrives.
- The system shall apply a cancellation fee if the ride is canceled after the driver has arrived
- Refund requests shall be provided when sending a complaint in case of a valid complaint.
- Riders and drivers shall receive notifications of cancellations.

Nonfunctional:

- Refund requests shall be processed within 24 hours.
- Cancellation fees shall be automatically calculated based on distance

3.8 multistop rides

Functional:

- Before confirming their rides, riders shall be able to include up to three stops.
- The fare shall adjust automatically based on the additional stops made.
- The driver shall receive real-time routing updates with each individual stop, if more than one stop is set.
- Users shall be able to change or remove stops before the start of the ride.

Nonfunctional:

- Extra stops can be changed until 5 minutes before the pickup.
- The calculation of fare shall be done in 3 seconds after the stops have been changed

3.9 Customer support and help center (safety office)

Functional:

- Users shall be able to call AUC safety office in case of emergency.
- A FAQ section shall be available for common issues.
- Admins shall be able to track pending support tickets and update users on resolutions.

Nonfunctional:

- Complaint tickets shall be reviewed within 48 hours.

4.0 general

Functional:
mobile application shall be created in Flutter for cross platform compatibility.

the backend shall use Supabase with Node.js to allow for real time changes,

User activity shall be recorded for 6 months for security purposes.

Ride and user activity analytics shall be made available to admins.

Nonfunctional:
The response time for core actions shall be under 5 seconds.
Servers should have a minimum of 95% uptime.

Identified Risks

1. **AUC Cooperation:** The app requires integration with AUC's database for authentication, which could be challenging to coordinate.
2. **User Adoption:** Convincing students to adopt the app and use it regularly may be difficult.
3. **Safety Concerns:** Ensuring the safety of riders and drivers, especially when sharing personal information like phone numbers and locations.
4. **Payment Issues:** Integrating multiple payment methods and ensuring secure transactions.
5. **Penalty System:** Implementing and enforcing penalties for late arrivals may be difficult and could lead to disputes between users or users refusing to use the app.